

7.9 (7.81) gives us the posterior distribution for the weights in regression RVM.

The product of all the priors of the elements of $\mathbf{w}$ is:

$$\begin{aligned} & \prod_{i=1}^M \mathcal{N}(w_i | 0, \alpha_i^{-1}) \\ &= \prod_{i=1}^M \frac{1}{\sqrt{2\pi\alpha_i^{-1}}} \exp\left(-\frac{w_i^2}{2\alpha_i^{-1}}\right) \\ &= \frac{\prod_{i=1}^M \sqrt{\alpha_i}}{\sqrt{(2\pi)^M}} \exp\left(-\frac{1}{2} \left(\sum_{i=1}^M \alpha_i w_i^2\right)\right) \end{aligned}$$

Let $\mathbf{A}$ be a diagonal matrix such that:

$$\mathbf{A} = \begin{bmatrix} \alpha_1 & 0 & \dots & 0 \\ 0 & \alpha_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \alpha_M \end{bmatrix}$$

Then,

$$\sum_{i=1}^M \alpha_i w_i^2 = \mathbf{w}^T \mathbf{A} \mathbf{w}$$

and since determinant of a diagonal matrix is the product of all it's diagonal values,

$$\begin{aligned} & \prod_{i=1}^M \sqrt{\alpha_i} = \sqrt{|\mathbf{A}|} \\ \Rightarrow \prod_{i=1}^M \mathcal{N}(w_i | 0, \alpha_i^{-1}) &= \frac{1}{\sqrt{(2\pi)^M |\mathbf{A}|^{-1}}} \exp\left(-\frac{1}{2} \mathbf{w}^T \mathbf{A} \mathbf{w}\right) \\ &= \mathcal{N}(\mathbf{w} | \mathbf{0}, \mathbf{A}^{-1}) \end{aligned}$$

The conditional distribution of the target variable given by (7.76) is of the same form as (3.8), which in turn gives us a likelihood function of the same form as in (3.10). Since we have obtained a prior over $\mathbf{w}$ of the same form as (3.48), where:

$$\mathbf{m}_0 = \mathbf{0}$$

$$\mathbf{S}_0 = \mathbf{A}^{-1}$$

we can substitute these into (3.50) and (3.51), and we get:

$$\mathbf{m} = \Sigma(\mathbf{A}\mathbf{0} + \beta\Phi^T\mathbf{t})$$

$$= \beta\Sigma\Phi^T\mathbf{t}$$

and

$$\Sigma = \left(\mathbf{A} + \beta\Phi^T\Phi\right)^{-1}$$